

SCC.160 – Fundamentals of Communication Systems

Tutorial Questions and Revision (Reliable Transmission of Information)

Knowledge

- Q1.** How does a single-bit error differ from a burst error?
- Q2.** Discuss the concept of redundancy in error detection and correction.
- Q3.** Distinguish between forward error correction versus error correction by retransmission.
- Q4.** What is the definition of a linear block code? What is the definition of a cyclic code?
- Q5.** What is the Hamming distance? What is the minimum Hamming distance?
- Q6.** How is the simple parity check related to the two-dimensional parity check?
- Q7.** What kind of arithmetic is used to add data items in checksum calculation?
- Q8.** What kind of error is undetectable by the checksum?
- Q9.** List the **three main operations** that can be performed on a message (e.g., a file, an image, a document, etc). How do they affect the size of the message?
- Q10.** How do **channel coding** and **modulation** affect data reliability and data rate (throughput)?

Understanding

Q1. What is the maximum effect of a 2-ms burst of noise on data transmitted at the following rates?

- a. 1500 bps
- b. 12 kbps
- c. 100 kbps
- d. 100 Mbps

Q2. Exclusive-OR (XOR) is one of the most used operations in the calculation of codewords. Apply the exclusive-OR operation on the following pairs of patterns. Interpret the results.

- a. $(10001) \oplus (10001)$
- b. $(11100) \oplus (00000)$
- c. $(10011) \oplus (11111)$

Q3. Using the code in Table below, what is the dataword if each of the following codewords is received?

- a. 01011
- b. 11111
- c. 00000
- d. 11011

<i>Dataword</i>	<i>Codeword</i>	<i>Dataword</i>	<i>Codeword</i>
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

Q4. What is the Hamming distance for each of the following codewords?

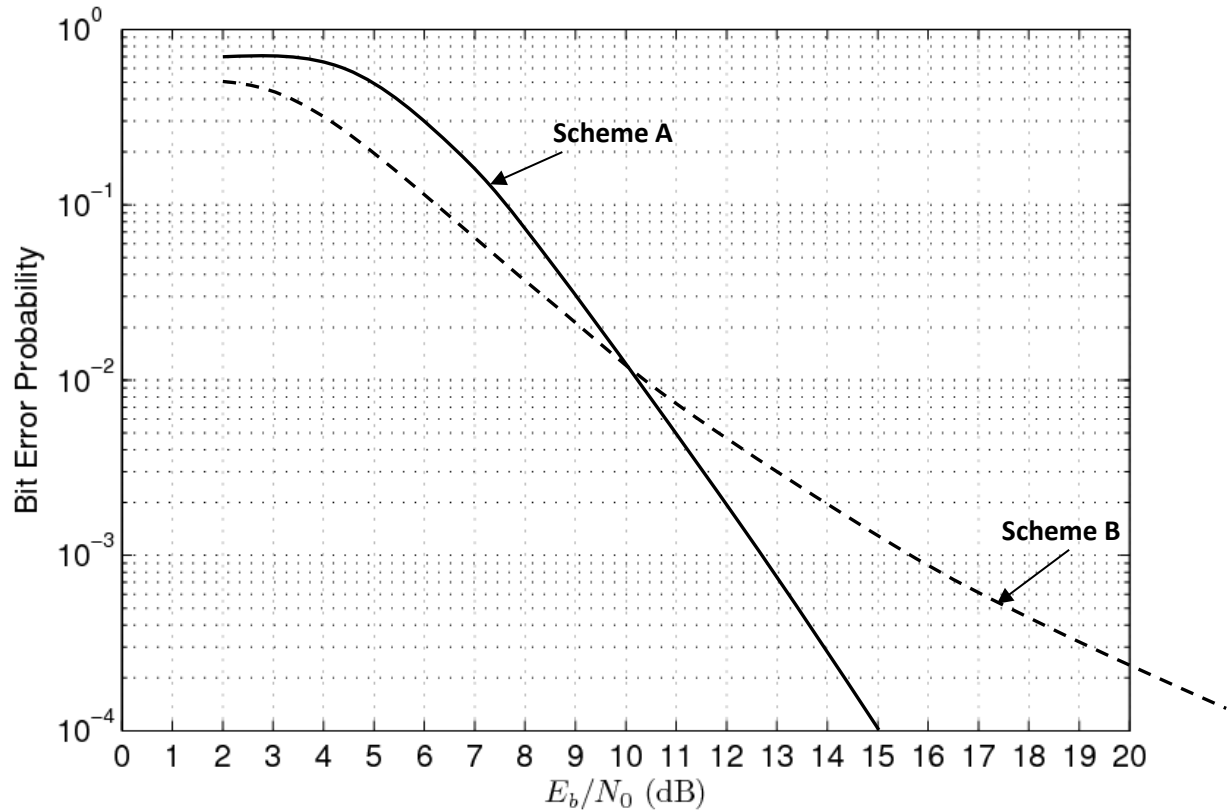
- a. $d(10000, 00000)$
- b. $d(10101, 10000)$
- c. $d(00000, 11111)$
- d. $d(00000, 00000)$

Q5. Assuming even parity, find the parity bit for each of the following data units.

- a. 1001011
- b. 0001100
- c. 1000000
- d. 1110111

Q6. Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using binary division).

Q7. The performance of **two** transmission schemes (A and B) is depicted in the figure below. Answer the following questions:



- Which scheme is **better**?
- If we transmit 5000 bits using Scheme B at $E_b/N_0 = 15$ dB, how many **bits** should we expect to receive in **error**?
- In order to achieve a bit error probability of 5×10^{-3} using Scheme A, how **many times** should the energy per bit E_b be higher than the power spectral density N_0 ?
- Assume that Scheme A corresponds to an **uncoded modulation** scheme that uses **8 symbols**. What is the required **SNR** (in dB) to achieve a bit error probability of 10^{-4} ?